Federated Data Platform - 2022

Supplier Briefing Call, Wednesday 13 April 14:00

Better insights. Better decisions. Better health.

Introductions

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Agenda

1. Background and purpose (JR)
2. Context (MT)
3. Projected benefits (MT)
4. Specification and Use Cases (AB)
5. Overview of procurement (KC)
6. RFI supplier questionnaire (KC)
7. Next steps (JR)
8. Q&A (JR)

Background and purpose

The planned Federated Data Platform (FDP) is an integrated data platform that will be owned and controlled by the NHS to unlock the value of NHS data.

It is estimated that an effective FDP has the potential to deliver £3.6bn benefits over 10 years.

The purpose of this briefing call is to give a broad overview of the planned FDP procurement, provide the opportunity to ask immediate questions on the market engagement, and advise of next steps and proposed procurement timelines.



Context – The Authority

The National Health Service Commissioning Board, commonly known as NHS England is an executive non-departmental public body of the Department of Health.

NHS England's key roles are to:

- ✓ allocate resources to Clinical Commissioning Groups and emerging ICS' to support them to commission services on behalf of their patients, according to evidence-based quality standards;
- ✓ directly commission primary care, military and offender health and specialised services including high security psychiatric services;
- ✓ take autonomous decisions about how best to allocate commissioning resources, remaining accountable for ensuring expenditure remains within limits set by the Secretary of State for Health;
- ✓ focus on achieving equal access to health services, designed around the needs of the patient, for which we will be held to account by Ministers and Parliament; and
- ✓ deliver improved patient outcomes.

Objectives of the Federated Data Platform

- Wider use of intelligent data and insights will be central to NHS recovery and, will address health inequalities and encourage innovation.

Objectives of the Population Health and Data Platform

Driving operational efficiency and effectiveness, providing access for decision-makers to accurate, real-time information to better understand and anticipate demand for services. Reducing the industry of local data manipulation and duplication of effort

Federating data so it's available to all users for appropriate use cases for an operating picture that is consistent, accurate and verifiable at the most appropriate level of granularity for decision making. Delivered by creating the infrastructure to separate data from capabilities

Moving to 'on demand' data and reporting products, reducing data latency so that up to date data is able to be used to provide insights for operational decision-making when it is needed. Ensuring that when data is wanted its there.

Providing the connectivity within the platform to enable local systems to better understand their capacity and operational service impacts, so they can respond to external changes and optimise health and care services for their population

Providing the operational flexibility for national regional and ICS teams to develop tools to optimise allocation and investments based on a common operating picture of health and care services

Maximising investment and realising the benefits early by working with suppliers & stakeholders to deliver the vision, utilising Agile best practice to delivery a MVP rapidly, enabling early benefits delivery

Reducing reporting burdens whilst increasing interoperability of systems to support delivery of high quality services, in turn improving operational efficiency and effectiveness of staff

Coordinating operational care and research with evidence-based processes and initiatives that grow the reputation, collaboration and knowledge base of the NHS and wider health and care system

Context – Why, What, How

- NHS England intends to go to market to procure a Federated Data Platform which will be an essential enabler to transformational improvements across the NHS.
- This brief summarises why such a FDP is needed, what it will deliver and how it will be procured, adopted and implemented.

Why

The NHS has demonstrated through the pandemic how data tools are critical to transforming operations.

- Providing frontline staff the information they need to provide the best possible care for patients.
- Securing insights to put patients, citizens and population health at the centre of service design.
- Enabling ICSs to effectively coordinate care.
- Providing the right people with the right access to information for strategic and operational planning.
- Increasing transparency and patient understanding of how data is used positively.

An effective FDP has the potential to deliver £3.6bn benefits over 10 years.

What

The FDP will be an ecosystem of technologies and services implemented across the NHS in England:

- The FDP itself, with ICS integration and consultancy and communications support for ICS implementation and adoption
- Privacy-enhancing technology
- Marketplace for applications
- Training and deployment support

The FDP will be built on five use cases:

- Population health and person insight; Care coordination (ICS); Elective recovery (Trust); Vaccines and immunisation; Supply Chain.

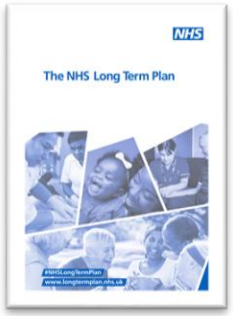
How

A procurement will be run for the FDP and ecosystem to ensure NHS realises best value for money.

- A robust communications approach will be implemented to maximise the opportunities with Systems and Providers.
- Implementation will not be forced across ICS teams – implementation approach will allow for a configurable adoption pattern where provided systems can meet standards
- The solution can be implemented in part or in whole and with an equitable distribution of implementation support.

The Federated Data Platform is aligned with NHS and Government strategic priorities

Why | What | How



NHS Long Term Plan

- *Fully integrated community-based health care supported by multidisciplinary teams*
- *Wider action on prevention and tackling health inequalities*
- *Increased use of data and predictive techniques to support local ICSs to plan and optimise care*
- *Interoperable digital services*
- *1.1% annual efficiency and productivity target*



Elective Recovery Plan

- *Rethink NHS delivery of screening, treatment and operations with clearer communications on timescales.*
- *Enable patients more control over their health and offer alternative treatments to reduce waiting times.*
- *Demonstrate the value of innovative data and technology solutions to free up staff time*
- *Recognise the importance of inclusive recovery*



Trusted Research Environment

- *Supporting TRE's to manage information security within their environments using defined processes, people and technology*
- *The FDP will support re-identification and de-identification, and deploying and configuring workflows, functions and tools in a privacy-enhancing manner*



Health and Social Care White Paper 2021

- *Development of ICSs to support joint-working between the NHS, local government and other partners in subsystem localities*
- *Enable local systems to understand capacity and operational impact.*
- *Provide operational flexibility for national, regional and ICS teams to optimise allocation and investments based on a common operating picture of health and care services*



NHS Data Strategy

- *Build on the success of data usage in the COVID-19 pandemic.*
- *Continue to use data to save lives and reshape health and care.*
- *Improve access data to generate insights, evidence and understanding to inform decision making*
- *Modernise architecture and ensure separation of data and applications.*



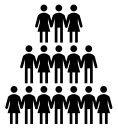
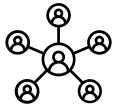
£200 million to boost healthcare research

- *£200 million committed by BEIS & DHSC to support NHS-led health research into diagnostics and treatment through new privacy-preserving platforms and clinical research services*
- *Funding will support the Life Sciences Vision and Government's levelling up agenda by supporting the creation of highly skilled jobs and backing world-leading health research*

The FDP will also enable key Government priorities including ***improved health services***, ***levelling up*** *and* ***pandemic preparedness.***

NHS England has defined five target outcomes in support of an overarching vision

Why | What | How

Vision	Connecting the NHS to transform care and improve outcomes for patients				
	1 	2 	3 	4 	5 
Target outcomes	Frontline staff have the information they need to provide the best possible care for patients	Population Health management - putting patients and citizens at the centre of service design	ICSs are able to effectively coordinate care	The NHS has access to the information it needs to undertake strategic and operational planning	There is increased transparency and patient understanding of how data is used positively
What it will mean	*"I have access to the information I need – in one place – enabling me to deliver the most appropriate care for each patient"*	*"I have the insights I need to proactively plan services around the peoples needs – which they can access when and where they need them"*	*"I have the information I need to proactively coordinate care"*	*"I understand where the NHS needs to invest in workforce and capacity, and can negotiate better deals"*	*"I support and trust where my data is being used for my care and the benefits of others"*

Projected FDP Benefits

Why | What | How

- The Federated Data Platform will enable programmes of change that will incrementally provide benefits on some of the highest areas of opportunity across the NHS.
- We expect to specify further savings – primarily linked to population health. Benefits will be tracked in conjunction with other programmes where benefits will be enabled by the FDP.

Population Health

- *Utilisation of individual-level data that provides linkages around a number of different datasets will allow clinicians to make more targeted and informed decisions around patient care.*
- *Further, timely data will allow for quicker and more proactive interventions, improving patient outcomes, targeting inequalities and reducing overall treatment costs.*
- *E.g. vaccination and screening*

Care Coordination

- *Improved access to patient-level data will provide commissioners and managers with a higher degree of real-time evidence upon which to coordinate care.*
- *This will help to ensure patients needs are met and information about care is shared to maximise benefits for patients.*
- *E.g. Waiting list and prioritisation*

Service Delivery

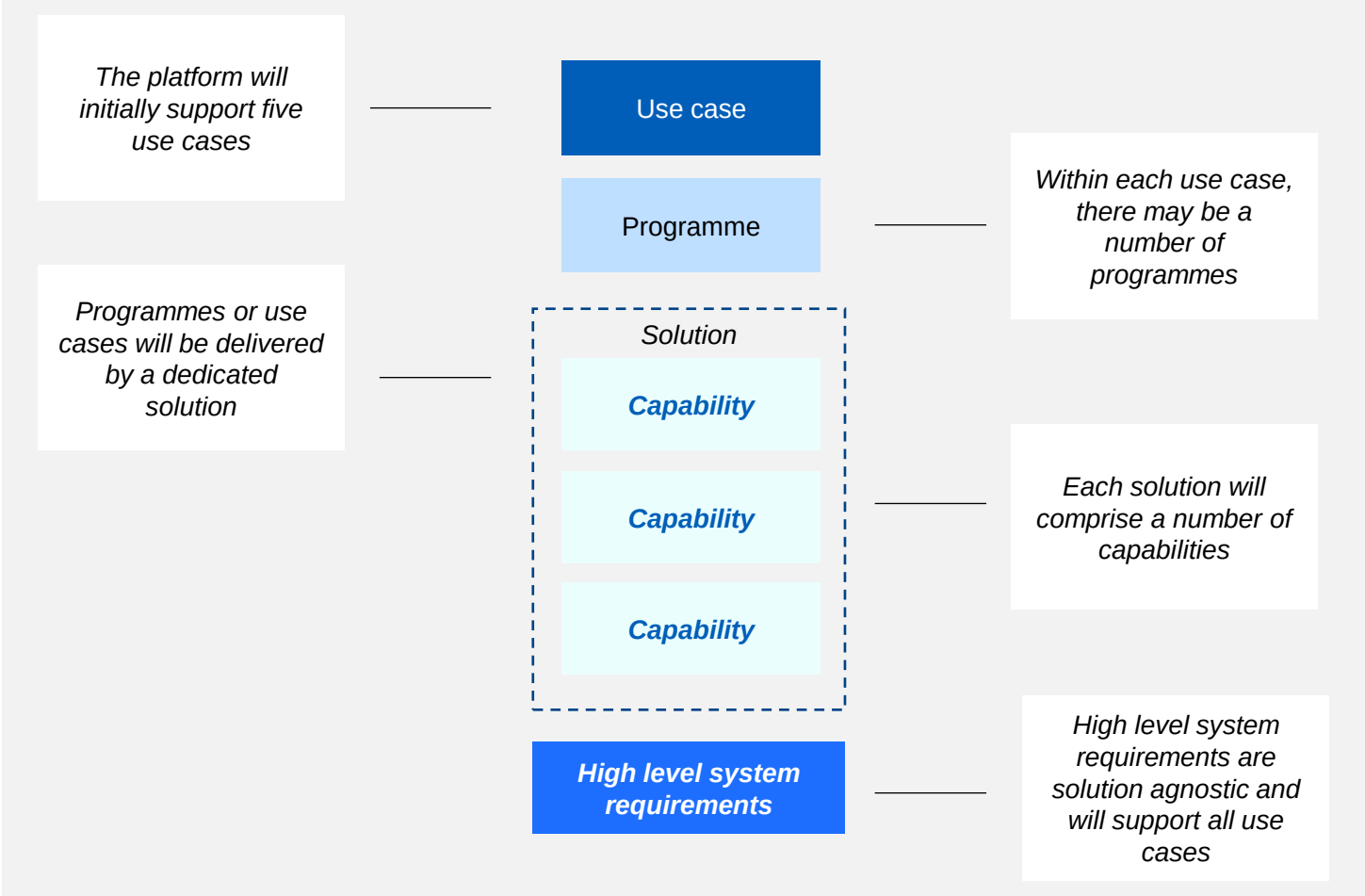
- *Individual trusts and other providers will be able to use the FDP to improve the efficiency of their workforce, scheduling and delivery, helping to ensure more patients are seen at the right time and reducing negative outcomes while also providing a higher degree of sustainability for the system.*
- *E.g. Bed management*

Specification & Use Cases

FDP Taxonomy

Why | What | How

- We are prioritising and have defined five use cases.
- Within each use case, there may be a number of programmes.
- Each use case or programme is delivered by a solution, comprising a combination of capabilities.
- Capabilities will be reusable across multiple solutions supporting multiple programmes and use cases
- We have defined underpinning functional and non-functional high level system requirements – which are solution agnostic and will enable delivery of all use cases.
- **In this pack we will provide an overview of use cases and technical requirements.**
- **Further definition on capabilities will be provided at a later stage.**



The NHS is moving towards ‘four uses of data’ which are underpinning the FDP approach

Why | **What** | How

	Uses of data	Examples [alignment to aims/outcomes]	User Types	Access and Latency	Data Types
1	**Direct Care**	*E.g. safe cross-organisation care delivery (including beyond ICS boundary)* [improved patient safety, effectiveness, productivity, experience]	• Clinical and care staff • Citizens	• Real-time, read/write access • Identifiable data required	• **Linked event-based patient record** • Past/future appointments/pathway info • Plans, communications, safeguarding etc.
2	**Population health and proactive care**	*E.g. Screening/Prevention, Case finding, Proactive care, Decision Support, Outcomes evaluation* [improved population health/ effectiveness, reduced health inequality]	• Authorised health and care staff	• Daily / real time updates • Identifiable data required by authorised staff to enable direct care (ICS level users and below e.g. place, provider)	• **Aggregated linked patient records** (clinical, demographic, wider social determinants, ‘omic, outcomes) • Provider/Place/workforce data
3	**Planning, oversight and service improvement (NHS/care and non-NHS)**	*E.g. Capacity and demand management, health inequality insight, performance management (safety, quality, experience), workforce planning, financial MI/contracting* [improved patient safety, productivity]	• NHS/care: Authorised health and care staff incl. local authorities • Non-NHS: Public, other ALB users e.g. CQC, other gov depts	• Weekly -> hourly updates (depending on criticality of action) • Must not be identifiable • Must have clear principles for what level of sharing is appropriate to whom, including for open data / dashboards	• Operational data (capacity, demand, activity/utilisation, cost, experience, safety, assets) • **Aggregated linked patient records** • Aggregated linked workforce data
4	**Research and Innovation**	*E.g. clinical trial recruitment & follow-up, disease progression and understanding, deep-learning AI, trial set-up/sizing* [improved population health, reduced health inequality, economic impact]	• Academic, charity sector and industry researchers	• Weekly • Anonymised for majority of use cases • Trial recruitment and follow up requires re-identifiable data, imperfect but timely	• **Rich, linked, clinical and operational data over life course / clinical pathway** • Including ‘omic and patient reported data

Requirements for identifiability of patient/staff information and refresh rate differ depending on use and user type; access must be secure and auditable for all use cases

Despite disparate uses, the data types are rooted in linked person-level data

It is this common need for linked patient data that underpins our approach, reducing duplication and ensuring traceability of personal data

With the FDP we are initially focusing deployment on five use cases

Why | What | How

Category	Link to four uses of data	Use case	Description	Indicative capabilities
Population Health	1 2 3 4	Population health and person insight	• Providing data and segmentation tools to enable population health management locally • Linking between datasets to drive population insight	Segmentation models, population opportunity analysis based on need, population cohorting, variation in outcomes
	1 2 3 4	Vaccination and immunisation	• Real time information to support supply and workforce planning and operational decision making • Monitoring and supporting intervention around equality of immunisation and vaccination programmes	Medicines and equipment ordering, population cohorting, reporting for immunisation and screening
Care Coordination	1 3 4	Elective recovery (Trust)	• Improving workflow and transparency around waiting list management • Single tool to manage the backlog and maximise utilisation of existing capacity	Waiting list cleansing and management, theatre scheduling and optimisation, outpatient management, patient comms and scheduling, pre-op management
	1 3 4	Care coordination (ICS)	• Enabling ICSs to optimise end-to-end services for patients, understanding capacity across whole system • Focus on organisational interfaces (e.g. virtual wards, anticipatory care and discharges)	Anticipatory care, discharge coordination, care coordination centres, virtual wards
Service Delivery	3	Supply chain	• Aligning supply and demand • Increasing visibility of where stock is needed to optimise management of the supply chain • Spend analysis to drive value	Load balancing, supply chain data cleansing, supply chain management, medicines and equipment ordering

Key – alignment to four uses of data **1** *Direct care* **2** *Population health and proactive care* **3** *Planning, oversight and service improvement* **4** *Research and innovation – will be supported via link to TRE*

Mapping of use cases, programmes and capabilities

Why | What | How

Capability	Population Health Management	Care Coordination					Elective Recovery	Vaccination & Immunisation	Supply Chain
		Discharges	Anticipatory Care	Virtual Wards	Screening	Other (e.g. elective hubs)			
Brokering		✓							✓
Citizens invite	✓		✓		✓			✓	
Cohorting	✓		✓		✓		✓	✓	
○ Data cleansing	✓	✓	✓	✓	✓	✓	✓	✓	✓
○ Data enrichment	✓	✓	✓	✓	✓	✓	✓	✓	✓
Load balancing				✓		✓			
Medicines and equipment ordering				✓				✓	✓
Medicines supply chain								✓	✓
○ Forecasting, monitoring & evaluation	✓	✓	✓	✓	✓	✓	✓	✓	✓
Patient comms interface						✓	✓		
Pathway Management		✓	✓	✓	✓	✓	✓		
Remote monitoring interface			✓	✓					
Scheduling					✓	✓	✓		
Supply chain management								✓	✓

Use case | Programme | Capability | ○ Cross cutting capability

The FDP will support multiple adoption patterns to tailor to the needs of ICSs

Why | What | How

The adoption approach takes account of the individual differences in digital maturity of each ICS and their ability to achieve common standards:

- *Each ICS will be required to achieve common standards in data reporting and connectivity but can achieve these through a variety of different solutions.*
- *Tailoring the implementation to different adoption patterns allows a flexible and adaptable approach for each ICS no matter their current stage of digital development.*
- *ICSs will be provided with equitable access to key capabilities/ services to achieve connectivity.*
- *In all patterns we will be providing core data sets and national activity and reference data*
- *Through alignment with the shared care programme we will ensure there is a common standard for data integration.*

	Pattern 1 De-centralised		Pattern 2 De-centralised with common tools		Pattern 3 Federated with custom analytics		Pattern 4 Federated with nationally provisioned support	
Capabilities / Services	ICS Provisioned Infrastructure	Nationally Provisioned Infrastructure	ICS Provisioned Infrastructure	Nationally Provisioned Infrastructure	ICS Provisioned Infrastructure	Nationally Provisioned Infrastructure	ICS Provisioned Infrastructure	Nationally Provisioned Infrastructure
Data ingestion	X		X	X		X		X
Data storage	X		X	X		X		X
Indicative common Analytics services e.g. pathway mgt	X			X		X		X
Local analytics and innovation	X		X	X		X		
National Reporting	X			NA - automated		NA - automated		NA - automated
Research	X		X	X		X		X
Data sharing and augmentation		X		X		X		X

Further information on data flow between adoption patterns is included in Appendix B

Overview of approach to Procurement 1

Why | What | How

Procurement 1	Procurement 2	Procurement 3
Lot A: The Platform, out-of-the-box applications & ICS integration Lot B: Privacy-enhancing technology	Marketplace that enables application release and management	Training, deployment support and implementation

- It is anticipated that Procurement 1 will be procured using the open procedure.
- The intention of the procurement approach is to maximise competition, deliver flexibility and value for money.
- The estimated contract length is five years with a possible two-year extension.
- The estimated contract value for Procurement 1 is:
  - Lot A – est. £360m over five years
  - Lot B – est. £10m over five years
- Terms & Conditions are likely to be a modified version of the Model Services Contract.

Indicative procurement timetable

Why | What | How

Indicative Procurement Activity	Estimated timeframe
Pre-market engagement	1 April – 22 April 2022
Supplier briefing day	13 April 2022
Further supplier event (TBC)	May 2022
Develop suite of tender documents	April – July 2022
ITT Issued	July 2022
Anticipated tender period	6 weeks
Evaluation completed	September 2022
Anticipated contract start date	November 2022

RFI supplier questionnaire

Why | What | How

- The RFI and supplier questionnaire will help NHS England refine the requirements and better understand the market and potential solutions for FDP.
- No information provided by a supplier in response to the RFI will be carried forward, used or acknowledged in any way for the purpose of evaluating the supplier, in any subsequent formal procurement process.
- Suppliers will not be excluded from participating in the ITT if they do not respond to the RFI.
- Supplier Questionnaire Sections:
  - Basic supplier information
  - Requirements
  - Procurement process
  - Contract
  - Interest

Next steps

Why | What | How

Step	Summary	Indicative dates
Deadline for RFI responses	Interested suppliers to upload responses on Atamis portal	22 April 2022
Follow up briefing call (TBC)	Further to responses from RFI	May 2022
Launch procurement	ITT released to the market	July 2022

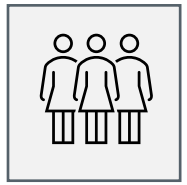
Questions?

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Appendix – Additional Information on Use Cases

Use cases
Population health and person insight

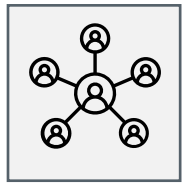
Use cases | High level requirements



Business problem	• National segmentation of person level data sets that are hosted on the NCDR often differs from the segmentation carried out at local levels, making benchmarking and patient specific service design challenging • Processes to generate system and place-based insight can be inconsistent, reducing the effectiveness of service design and resource allocation with respect to different patient cohorts.
Proposed solutions	• Enhanced processing and utilisation of data to inform cohorting of patients, enabling design of, and invitation of citizens into, specific services. • Monitoring and evaluation of population health initiatives enabling long term strategic investments in patient centred services. • Functionality to support a consistent approach to patient segmentation, risk stratification and impactability • Ability to add an additional level of detail and granularity to national segmentation by enriching with local level insights
Overall benefit	• Placing patients at the centre of service design, and providing timely access to insight needed to discover and identify patients. Thus planning services around patients needs that are well understood. • Providing data and segmentation tools to enable population health management locally, and linking between datasets to drive population insight. • Baseline segmentation for ICS to use and enrich locally. Bridging gaps and enabling comparison across organisations • Provide transparency regarding data use. Providing common methods, metrics and measures across organisations • Utilisation of individual-level data and linkages between datasets will allow clinicians to make targeted and informed decisions around patient care at a local level • Access to timely data will allow for quicker and more proactive interventions, improving patient outcomes, reducing inequalities and NHS treatment costs overall. • Using advanced analytics at a national level to provide insights on improving NHS inequalities across regions. To further improve the understanding of the impact of initiatives on public health outcomes.
Alignment with strategic priorities	• Wider action on prevention and tackling health inequalities. Linked data sets to support PHM – Long Term Plan • Improved access to data to generate insights, evidence and understanding to inform decision making – Data Strategy • Legal responsibility of ICS/ ICBs to report of pop health data- Health and Care Bill (white paper) • Improving patient care and the health of the population while reducing the cost of healthcare- The IHI Triple Aim

Use cases
Care coordination (1)

Use cases | High level requirements



Business problem

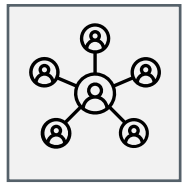
- Systems and providers often don't have the tools or infrastructure to enable sharing of patient level data (securely) across organisational boundaries. This reduces effectiveness of management across the patient pathway and the ability of an ICS to provide streamlined care for their populations.
- Unable to get a clear view of end-to-end care coordination insights: Data is disparate and in different structures, often with challenges to the quality and integrity. This causes data management challenges when linking the data from multiple sources to drive insight

Overall benefit

- Ensuring frontline staff have access to the information they need in one place to provide the best possible care for patients and enabling ICSs to proactively coordinate care.
- Enabling ICSs to optimise end-to-end services for patients, understanding capacity across the whole system, with a focus on organisational interfaces (e.g. virtual wards, anticipatory care, discharges, screening and elective hubs).
- Enhanced operational decision-making capability derived from a common operating picture that leads to better health outcomes
- Improved operational flexibility to respond to new business workflows and stand-up new operational functionality across organisational boundaries
- Ability to manage capacity across numerous providers maximise speed and quality of patient services
- Cost savings derived from ability to leverage local, regional and national data and analytics activities within a core platform and operating toolset
- Achieving a step change in demand and capacity modelling to enable cross provider patient pathway management
- Confidence in decision making / insights through strong data integrity and management/governance of data

Use cases
Care coordination (2)

Use cases | High level requirements



Indicative programmes and solutions

Discharges	Anticipatory Care	Virtual Wards	Screening	Other
• Enabling faster assessment and triage of patients to appropriate services upon discharge from acute care. • Effective brokering of system demand / supply based on 'real time' view of available resources, providing users with a choice of services to support patients and filling gaps in the pathway	• Identification of cohorts of patients and capability to reidentify and invite into specific services to ensure that the most appropriate care is delivered with the highest impact on patient outcomes. • Provide access to real time data to enable decision making to better coordinate care. Ensuring information regarding care is shared to maximise patient benefit. • Supporting collation of remote monitoring data.	• Assess, triage, and monitor patients using virtual wards. • Collation and integration of patient records for use by a multidisciplinary teams to coordinate care. • Access to remote monitoring data – IoT of patient reported – to inform clinical decision making.	• Support to local areas to run their screening processes. • Bringing together multiple datasets required to understand demand and capacity and coordinate screening	• Improved scheduling, pathway management and patient communication. • Standardisation, cleansing and enrichment of existing datasets with additional data as it is acquired to enable richer analysis to inform decision making. • Allow short, medium and long term planning to be conducted • Allow patients to take ownership of their data

The platform will provide access to real time data to enable decision making to better coordinate care. Ensuring information regarding care is shared to maximise patient benefit.

Alignment with strategic priorities

- Enable local systems to understand capacity and operational impact – **Health and Social Care White Paper 2021**
- Providing operational flexibility for national, regional and ICS teams to optimise allocation and investments based on a common operating picture of health and care services – **Health and Social Care White Paper 2021**

Use cases
Elective recovery

Use cases | High level requirements

Elective recovery



Business problem	• Unable to get a clear view of the Elective Recovery insights easily: Data is disparate, often with challenges to the quality and integrity, therefore causing data management challenges when linking the data from multiple sources to drive insight • Lack of local tooling to support effective operational processes such as scheduling
Proposed solutions	• Improved scheduling, pathway management and patient communication. • Standardisation, cleansing and enrichment of existing datasets with additional data as it is acquired to enable richer analysis to inform decision making. • Monitoring and evaluation of elective recovery initiatives and clinical outcomes • Allow short, medium and long term planning to be conducted
Overall benefit	• Standardisation, cleansing and enrichment of existing datasets with additional data as it is acquired to enable richer analysis to inform decision making. • Monitoring and evaluation of elective recovery initiatives and clinical outcomes • Providing short, medium and long term operational and strategic planning, improving efficiencies and outcomes • Better understanding of capacity and operational service impacts, enabling more effective prioritisation and use of capacity and capability to treat patients and reduce waiting times • A key enabler to inform system and regional care coordination centres on the current status of services • Minimise burden on the service by increasing the system wide supply of care in the system via mutual aid/ sharing capacity • Achieve higher operational productivity by providing information on activity and performance that streamlines payment flows across the system
Alignment with strategic priorities	• **Elective Recovery Plan** • Eliminate waits of over one year by March 2025, and waits of over two years by July 2022 • At least 95% of patients receiving tests within 6 weeks by March 2025 • Rethink NHS delivery of screening, treatment and operations with clearer communications on timescales. • Enable patients to have more control over their health and offer alternative treatments to reduce waiting times. • Demonstrate the value of innovative data and technology solutions to free up staffs time • Recognise the importance of inclusive recovery • Support ongoing delivery- **The NHS Constitution for England**

Use cases
Vaccinations and immunisations

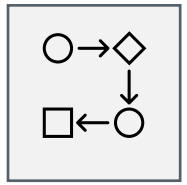
Use cases | High level requirements



Business problem	• A lack of real-time, detailed information on demand that is split by cohort across population • Lack of visibility regarding access and availability of vaccination and immunisation in order to ensure that preventative services are targeted towards the highest value cohorts of the population to maximise benefits
Proposed solutions	• Identification of cohorts of patients and capability to invite into targeted services. • Standardisation, cleansing and enrichment of existing datasets to enable richer analysis to inform decision making. • Improved visibility and understanding of medicines and vaccination demand and supply to inform supply chain and distribution. • Monitoring and evaluation to understand patient outcomes, inform programme performance, address inequalities and understand uptake.
Overall benefit	• Coordinating operational care and research with evidence-based processes and initiatives that grow the reputation, collaboration and knowledge-base of the NHS and the wider health and care system. E.g. inform publications using statistical and reporting suites • Enabling real time information to support supply and workforce planning and operational decision making • Monitoring and supporting interventions around equality of immunisation and vaccination programmes • Better understanding of demand by cohort and of access and availability. • Real time monitoring of uptake and coverage to drive more timely decisions with confidence in our data • Reporting suites to: Improve programme performance & uptake and ensure immunisation programmes are delivered equitably • Enable the broader application of the COVID vaccines infrastructure to support S7a services
Alignment with strategic priorities	• Pandemic preparedness: Develop upon the success of the Foundry and ensure that England is well equipped and prepared for future pandemics – HMG • Build on the success of data usage in the COVID-19 pandemic – NHS Data Strategy • Continue to use data to save lives and reshape health and care – NHS Data Strategy

Use cases
Supply chain

Use cases | High level requirements



Business problem	• There is inefficiency in current supply chain management processes with data held in silos leading to a lack of visibility of current processes. There is also a lack of coordination nationally and locally and an inability to match supply with demand
Overall benefit	• Increasing visibility of where stock is needed to optimise management of the supply chain • Reducing resource constraints by enabling proactive management and increased visibility of the end-to-end supply chain. • Improved visibility and understanding of demand and supply enabling optimised management of the supply chain and enhanced spend analysis to drive value of procurement. • Ensure effective investment and capital efficiency • Ensure equality of services across all systems, delivering resources to the areas of greatest need
Proposed solutions	• Standardisation, cleansing and enrichment of existing datasets with additional data as it is acquired to enable richer analysis to inform decision making. • Providing operational workflow support monitoring of processing orders. Delivering and distributing supply across the system • Understand rundown rates, monitor live utilisation and support auto replenishment • Allow benchmarking of products based on features such as cost, time scales and quality
Alignment with strategic priorities	• Improve access data to generate insights, evidence and understanding to inform decision making – Data Strategy. • 1.1% annual efficiency target – Long Term Plan • Greener NHS- Delivering a net zero NHS